FAI-F017CA3-1

UNITED STATES
DEPARTMENT OF LABOR
MINE SAFETY AND HEALTH ADMINISTRATION

REPORT OF INVESTIGATION

Surface
(Construction Sand and Gravel)

Fatal Powered Haulage Accident
March 28, 2025

South Plant
Midwest Materials By Mueller Inc
Hanover, Washington County, Kansas
ID No. 14-01414

Accident Investigators

Christopher Ewing
Mine Safety and Health Specialist

Kelly Burnham
Supervisory Mine Safety and Health Inspector

Originating Office
Mine Safety and Health Administration
Madisonville District
100 YMCA Drive
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Mary Jo Bishop, District Manager

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OVERVIEW

On March 28, 2025, between 4:00 p.m. and 4:30 p.m., Daniel Holle (D. Holle), a 51-year-old front-end loader operator with 28 years of mining experience, died when sand from the highwall engulfed the front-end loader he was operating.

The accident occurred because the mine operator did not: 1) examine ground conditions as warranted during the work shift, and 2) use mining methods that maintained highwall stability.

GENERAL INFORMATION

Midwest Materials By Mueller Inc owns and operates the South Plant mine. The mine is a surface construction sand and gravel mine located near Hanover, Washington County, Kansas. The mine employs three miners and operates one ten-hour shift, five days per week. The mine operator uses an excavator to remove overburden (dirt and clay), to reach sand and gravel from the highwall. Front-end loaders stockpile the sand and gravel from the highwall and load it into haul trucks. Haul trucks transport the sand and gravel to the plant for sizing. Once sized, the sand and gravel are loaded into customer trucks for commerce and for use at a ready-mix concrete plant.

The principal management officials at the South Plant mine at the time of the accident were:

Robert Holle	Owner/President
Darwin Frank	Superintendent

The Mine Safety and Health Administration (MSHA) completed the last regular safety and health inspection at this mine on March 13, 2025. The 2024 non-fatal days lost incident rate for the South Plant mine was zero, compared to the national average of 0.97 for mines of this type.

DESCRIPTION OF THE ACCIDENT

On March 28, 2025, at approximately 7:00 a.m., D. Holle started his shift and met with Robert Holle (R. Holle), President/Owner, at a ready-mix concrete shop close to the mine, also owned by R. Holle. At 7:25 a.m., D. Holle and R. Holle traveled to a grading project eight miles away. At 11:00 a.m., R. Holle instructed D. Holle to return to the mine to stockpile sand.

At approximately 12:30 p.m., Darwin Frank, Superintendent, saw D. Holle arrive at the plant, fuel up the front-end loader, and then drive it into the pit. At 3:30 p.m., Frank drove to the pit and saw D. Holle using the front-end loader to stockpile sand near the base of the highwall. At 4:00 p.m., Frank returned to the plant. At 4:32 p.m., Frank sent D. Holle a text message but did not receive a response, so he drove into the pit. Frank found the front-end loader covered in sand. He ran up to the front-end loader, yelled out for D. Holle, but did not receive a response. At 4:42 p.m., Frank called 911. Frank also called R. Holle.

At approximately 5:00 p.m., Washington County Sheriff's Department, Washington County Fire Department (FD), and the City of Washington Emergency Medical Services (EMS) arrived. At 9:10 p.m., EMS recovered D. Holle from the front-end loader. Kellen Sherlock, M.D. from Washington County Hospital, pronounced D. Holle dead at 9:28 p.m.

INVESTIGATION OF THE ACCIDENT

On March 28, 2025, at 6:34 p.m., Joel Curtiss, Fire Chief for Washington County FD, called the Department of Labor National Contact Center (DOLNCC). The DOLNCC contacted Abel DeLeon, Supervisory Mine Safety and Health Inspector, who contacted Matthew Stone, Acting Assistant District Manager. Stone sent Kelly Burnham, Supervisory Mine Safety and Health Inspector, and Christopher Ewing, Mine Safety and Health Specialist, to the mine. Stone assigned Ewing as the lead investigator.

On March 29, 2025, at 12:00 a.m., Ewing arrived at the mine and issued an order under the provisions of Section 103(k) of the Mine Act to ensure the safety of the miners and the preservation of evidence. On March 29, 2025, at 8:30 a.m., Ewing returned, and Burnham arrived at the mine to continue the investigation.

The MSHA accident investigation team conducted an examination of the accident scene, interviewed mine management, and reviewed conditions and work procedures relevant to the accident. MSHA Technical Support performed a stability analysis of the highwall and provided recommendations for the mining method in the corrective actions. See Appendix A for a list of persons who participated in the investigation.

DISCUSSION

Location of the Accident

The accident occurred at the base of the 55 to 60-foot highwall in the only active pit at this mine (see Appendix B).

Weather
The weather at the time of the accident was 74 degrees Fahrenheit and sunny with 20 mile per hour winds. Investigators determined that the weather did not contribute to the accident.

Equipment Involved
The equipment involved in the accident was a Volvo L120E front-end loader. The force of the sand broke the front window, allowing sand to enter the operator's cab. The windows were original to the front-end loader and had no defects. Due to the damage caused by the accident, the investigators were unable to determine if the front-end loader had any safety defects prior to the accident.

Mining Method
At the time of the accident, D. Holle was removing sand from a 14-foot-wide area at the base of the highwall. The 55 to 60-foot highwall was severely undercut. D. Holle had been digging at the base of the highwall at a downward trajectory on approximately a five percent grade. D. Holle stockpiled the sand on the outer corner of the highwall (see Appendix C). The stockpile was approximately 50 feet wide and the edge closest to the highwall was approximately 50 feet away from the base. According to interviews, this was a normal mining method at the mine, but not the primary mining method. The mine operator's primary mining method was to use an excavator that worked from an elevated pad at the base of the highwall. The excavator maintained a trench between the highwall and the elevated pad to catch falling sand. The mine operator did not have any written procedures for either of these mining methods. Investigators determined that neither mining method is safe for the type of material and height of the highwall at this mine. The removal of sand from the base of the highwall created hazardous ground conditions. The mine operator did not use mining methods that maintained highwall stability, which contributed to the accident.

Examinations
Frank conducted a pre-operational inspection of the Volvo L120E front-end loader on the day of the accident and did not identify any defects. Investigators determined the pre-operational inspection was adequate and did not contribute to the accident.

Frank conducted a workplace examination of the pit between 8:00 a.m. and 9:30 a.m. on the day of the accident and did not identify any hazards. Investigators determined the workplace examination was adequate and did not contribute to the accident.

Frank conducted a ground condition examination in conjunction with the workplace examination and did not identify any hazards. The highwall became severely undercut while D. Holle was removing sand from the base of the highwall, and ground conditions changed with every load of sand removed. An adequate ground condition examination of the highwall as the conditions changed during the shift would have identified that the undercut material created an unstable highwall. Frank saw D. Holle using the front-end loader to stockpile sand near the base of the highwall and there was no excavator in the pit, indicating that D. Holle was removing sand from the base of the highwall with the front-end loader. The mine operator did not examine ground conditions as warranted during the work shift, which contributed to the accident.

Training and Experience

D. Holle had 28 years of mining experience, all at the South Plant mine. D. Holle received site-specific hazard awareness training and task training on a front-end loader on February 14, 2008, and annual refresher training on February 4, 2025. The site-specific hazard awareness training and annual refresher training included identification of hazardous ground conditions. The mine operator could not provide documentation for D. Holle's task training on the Volvo L120E front-end loader or either mining method used at the mine. Due to D. Holle's experience operating other front-end loaders at this mine, investigators determined that the lack of task training did not contribute to the accident.

ROOT CAUSE ANALYSIS

The accident investigation team conducted an analysis to identify the underlying causes of the accident. The team identified the following root causes, and the mine operator implemented the corresponding corrective actions to prevent a recurrence.

1. Root Cause: The mine operator did not examine ground conditions as warranted during the work shift.

   Corrective Action: The mine operator developed and implemented a written procedure that requires ground condition examinations to be conducted prior to work beginning and as conditions warrant when adverse conditions are encountered. The mine operator trained all miners on the procedure.

2. Root Cause: The mine operator did not use mining methods that maintained highwall stability.

   Corrective Action: The mine operator developed and implemented a written procedure for a mining method that maintains highwall stability. The procedure requires: 1) the mine operator to maintain a 3:1 highwall slope, 2) equipment on top of the highwall to be at least 35 feet away from the edge of the highwall, and 3) vertical banks to be no higher than 25% of the height of the cab of the equipment being used to remove the sand. The mine operator trained all miners on the procedure.

CONCLUSION

On March 28, 2025, between 4:00 p.m. and 4:30 p.m., Daniel Holle, a 51-year-old front-end loader operator with 28 years of mining experience, died when sand from the highwall engulfed the front-end loader he was operating.

The accident occurred because the mine operator did not: 1) examine ground conditions as warranted during the work shift, and 2) use mining methods that maintained highwall stability.

Approved By:

_____ _____

Mary Jo Bishop
District Manager

Date

ENFORCEMENT ACTIONS

1. A 103(k) order was issued to Midwest Materials By Mueller, Inc.
   A fatal accident occurred on March 28, 2025, between 4:00 p.m. and 4:30 p.m. This order is being issued under the authority of the Federal Mine Safety and Health Act of 1977, under Section 103(k) to insure the safety of all persons at the mine and requires the operator to obtain the approval of an authorized representative of MSHA of any plan to recover any person in the mine or to recover the mine or affected area. This order prohibits any activity in the affected area. The operator is reminded of the obligation to preserve all evidence that would aid in investigating the cause or causes of the accident in accordance with 30 CFR 50.12.

2. A 104(d)(1) citation was issued to Midwest Materials By Mueller, Inc. for a violation of 30 CFR 56.3130.
   On March 28, 2025, a front-end loader operator died when sand from the highwall engulfed the front-end loader he was operating. The mine operator did not use mining methods that maintained highwall stability. The superintendent knew the front-end loader was working at the base of the highwall next to the stockpile area prior to the accident. The mine operator engaged in aggravated conduct constituting more than ordinary negligence. This violation is an unwarrantable failure to comply with a mandatory standard.

3. A 104(d)(1) order was issued to Midwest Materials By Mueller Inc for a violation of 30 CFR 56.3401.
   On March 28, 2025, a front-end loader operator died when sand from the highwall engulfed the front-end loader he was operating. The mine operator did not examine ground conditions as warranted during the work shift. The superintendent knew the front-end loader was working at the base of the highwall next to the stockpile area prior to the accident. The mine operator engaged in aggravated conduct constituting more than ordinary negligence. This violation is an unwarrantable failure to comply with a mandatory standard.

APPENDIX A – Persons Participating in the Investigation

Midwest Materials By Mueller Inc

Robert Holle	Owner/President
Darwin Frank	Superintendent

Mine Safety and Health Administration

Matthew Stone	Acting Assistant District Manager
Kelly Burnham	Supervisory Mine Safety and Health Inspector
Christopher Ewing	Mine Safety and Health Specialist
Brittany Horton	Mine Safety and Health Specialist
Jarrod Durig	Mine Waste and Geotechnical Engineering Division Chief, Technical Support
Darren Blank	Supervisory Civil Engineer, Technical Support

APPENDIX B – Aerial View



APPENDIX C – Highwall and Stockpile

